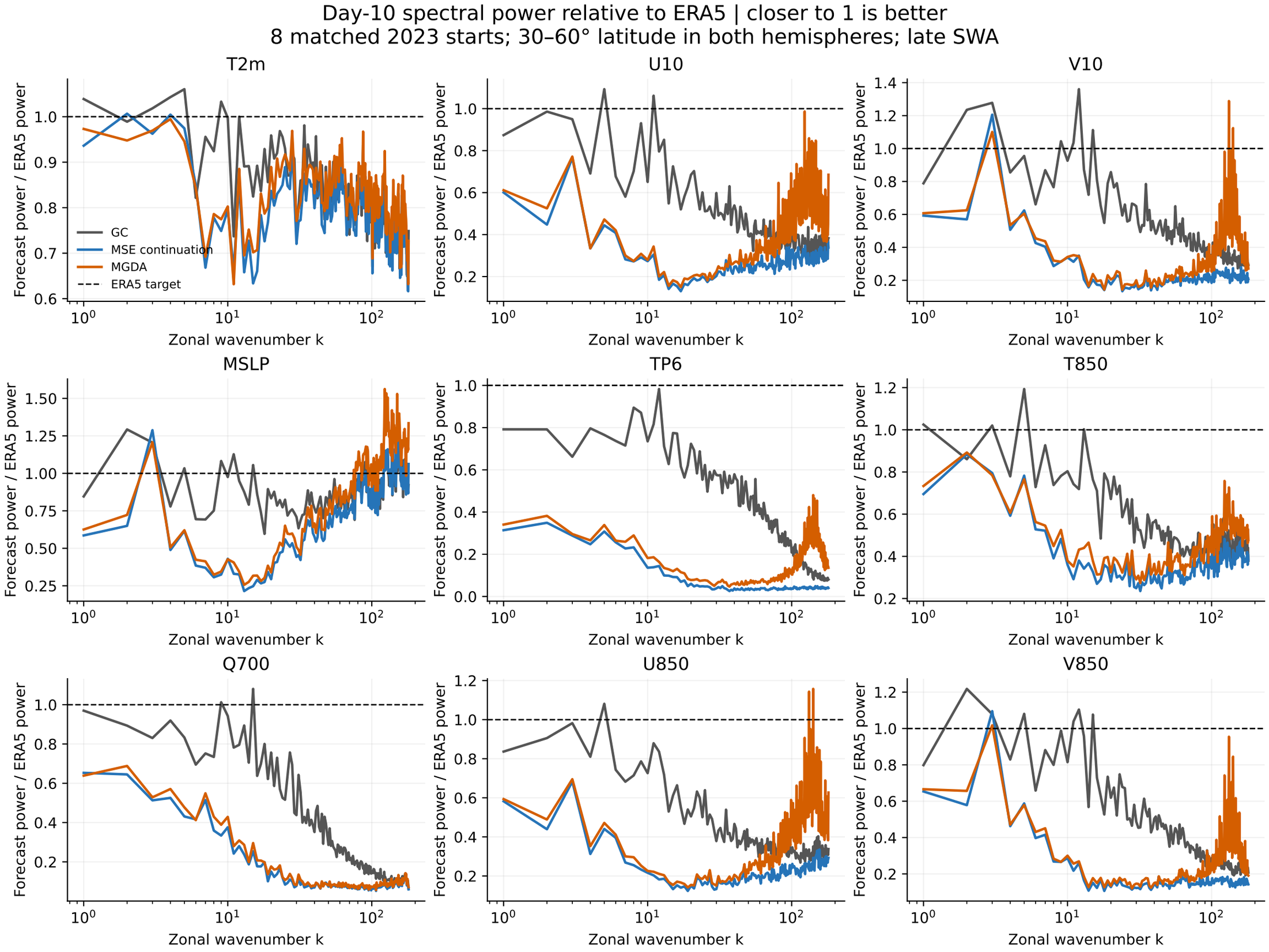
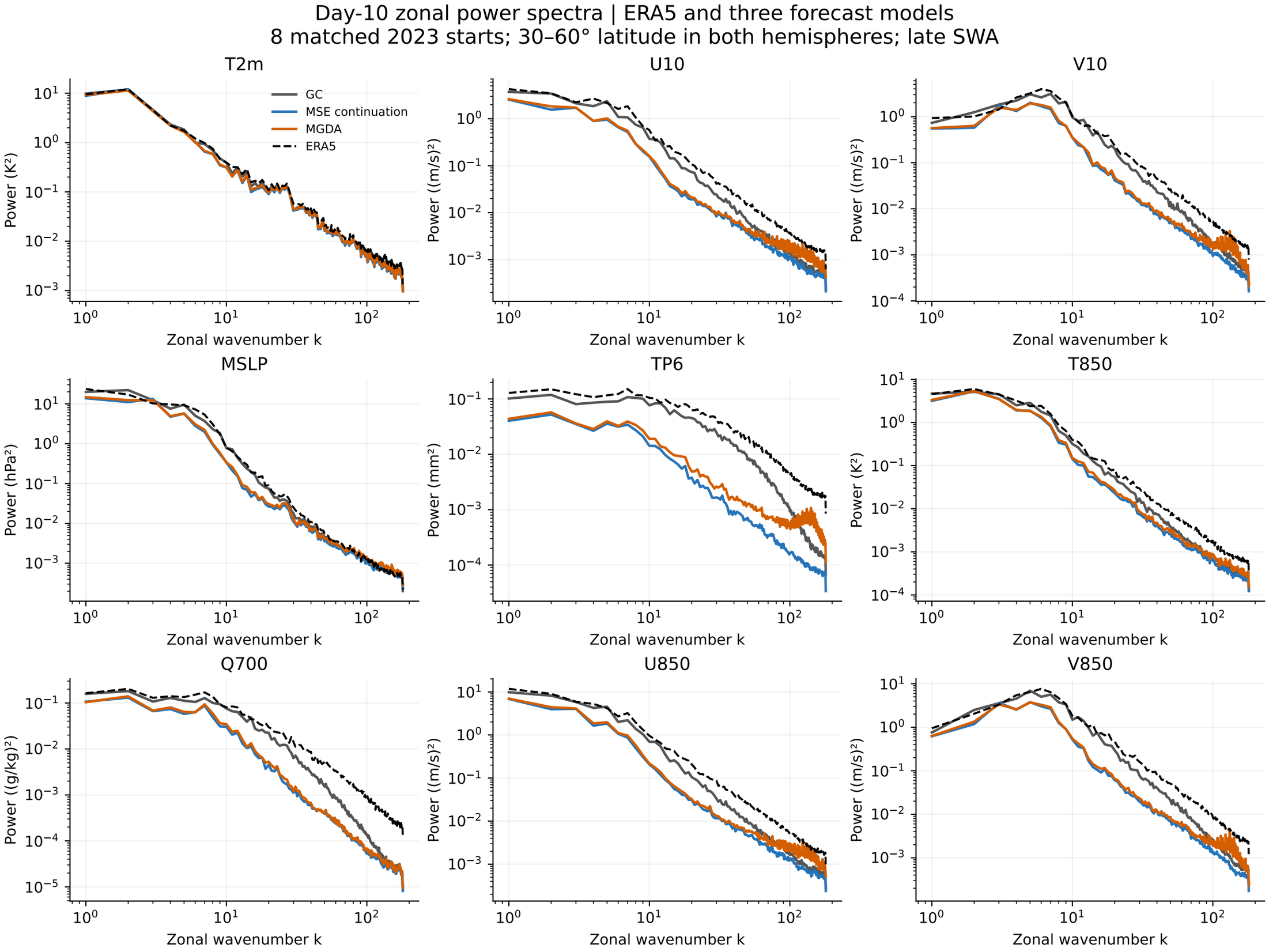
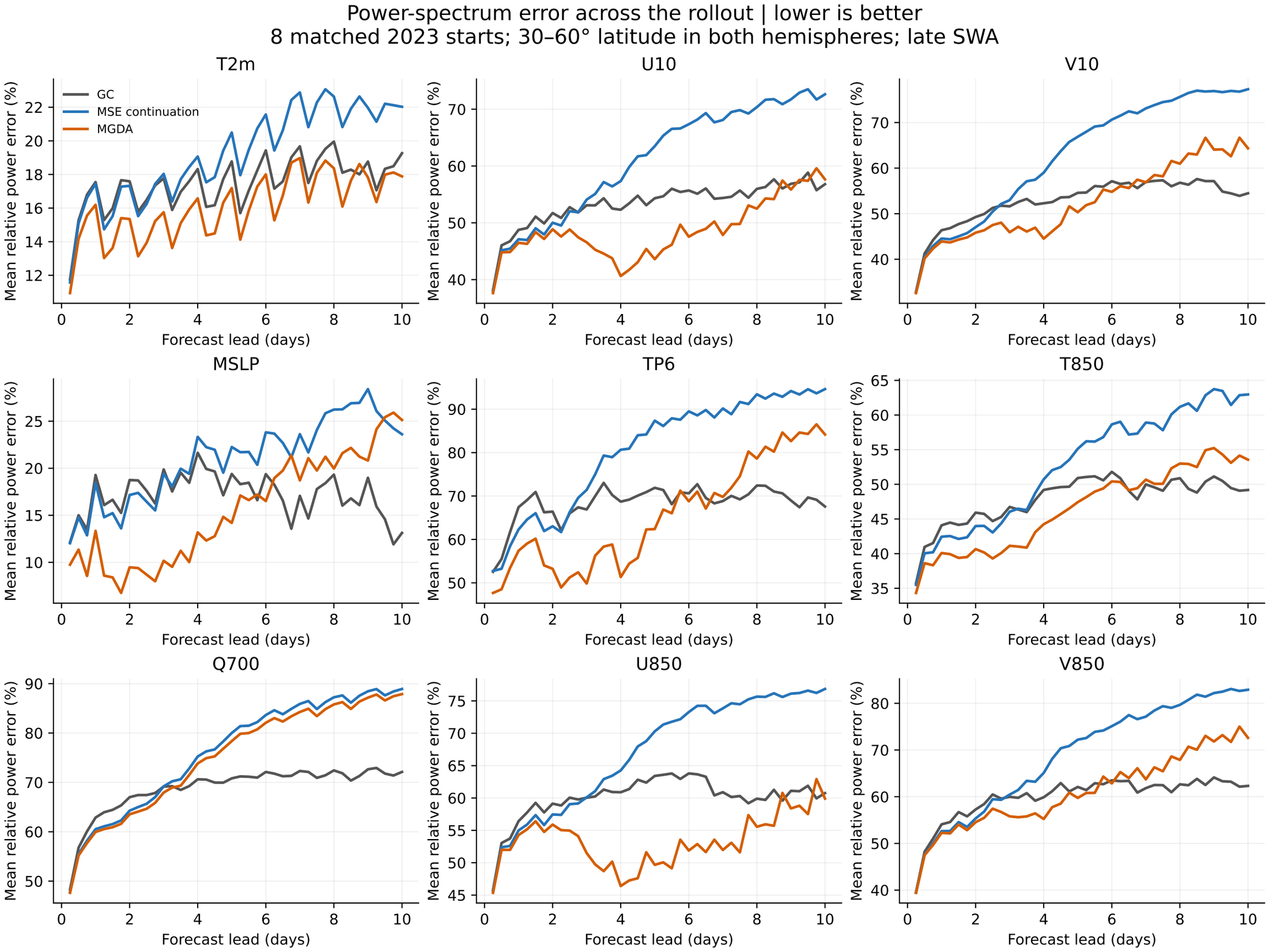

Day-10 spectral power relative to ERA5 | closer to 1 is better
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
Forecast power / ERA5 power
Zonal wavenumber k
GC
MSE continuation
MGDA
ERA5 target


Day-10 zonal power spectra | ERA5 and three forecast models
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
GC
MSE continuation
MGDA
ERA5
Power (K²)
Power ((m/s)²)
Power (hPa²)
Power (mm²)
Power ((g/kg)²)
Zonal wavenumber k


Power-spectrum error across the rollout | lower is better
8 matched 2023 starts; 30–60° latitude in both hemispheres; late SWA
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
GC
MSE continuation
MGDA
Mean relative power error (%)
Forecast lead (days)